

and for all n

$$\frac{\partial f(\phi)}{\partial \phi_n} = 0, \quad (27.6.18)$$

where the effective Kahler potential $K_\lambda(\phi, \phi^*)$ is

$$K_\lambda(\phi, \phi^*) = \mathcal{A}_\lambda(\phi, \phi^*, 0, 0 \cdots), \quad (27.6.19)$$

with $\mathcal{A}_\lambda(\phi, \phi^*, 0, 0 \cdots)$ obtained from $\mathcal{A}_\lambda$ by setting the gauge superfield and all superderivatives equal to zero. (With superderivatives required to vanish by Lorentz invariance, the only dependence of $\mathcal{A}_\lambda$ on V is a factor $\exp(-V)$ following every factor $\Phi^\dagger$.) We now use a trick that we have already employed in Section 27.4. If there is any solution $\phi^{(0)}$ of Eq. (27.6.18), then the gauge symmetry tells us that there is a continuum of such solutions, with ϕ_n replaced with

$$\phi_n(z) = \left[\exp(i \sum_A t_A z_A) \right]_{nm} \phi_m^{(0)}, \quad (27.6.20)$$

where (since f depends only on ϕ , not ϕ^*) the z_A are an arbitrary set of complex parameters. If $K_\lambda(\phi, \phi^*)$ has a stationary point anywhere on the surface $\phi = \phi(z)$, then at that point

$$0 = \sum_{nmA} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n} (t_A)_{nm} \phi_m \delta z_A - \sum_{nmA} \frac{\partial K_\lambda(\phi, \phi^*)}{\partial \phi_n^*} (t_A)_{mn} \phi_m^* \delta z_A^*. \quad (27.6.21)$$

Since this must be satisfied for all infinitesimal complex δz_A , the coefficients of both δz_A and δz_A^* must both vanish, and therefore Eq. (27.6.17) as well as Eq. (27.6.18) is satisfied at this point. Thus the existence of a stationary point of $K_\lambda(\phi, \phi^*)$ on the surface $\phi = \phi(z)$ would imply that supersymmetry is unbroken to all orders of perturbation theory. The zeroth-order Kahler potential $(\phi^\dagger \phi)$ is bounded below and goes to infinity as $\phi \rightarrow \infty$, so it certainly has a minimum on the surface $\phi = \phi(z)$, where of course it is stationary. If there were no flat directions in which K_λ is constant at this minimum then any sufficiently small perturbation to the Kahler potential might shift the minimum, but would not destroy it. At the minimum of the Kahler potential on the surface $\phi = \phi(z)$ there are flat directions: ordinary global gauge transformations $\delta \phi = i \sum_A \delta z_A t_A \phi$ with z_A real. But these are also flat directions for the perturbation $K_\lambda(\phi, \phi^*) - (\phi^\dagger \phi)$, so there is still a local minimum of K_λ on the surface $\phi = \phi(z)$ for any perturbation in at least a finite range, and thus to all orders in whatever couplings appear in $K_\lambda(\phi, \phi^*)$. As we have seen, this is a set of scalar field values at which $\mathcal{F}_n = 0$ and $D_A = 0$ for all n and A , which means that supersymmetry is unbroken.